

Kushagra Aggarwal

+91 9310834741 | kushagraaggarwal76@gmail.com

linkedin.com/in/kushagra-aggarwal | github.com/Kush-8912 | kushagra-aggarwal.vercel.app

Education

Scaler School of Technology

Full-time Computer Science & AI Program: 8.06/10.0

Aug. 2025 – Jul. 2029

Bengaluru, India

Birla Institute of Technology and Science, Pilani

Bachelor of Science in Computer Science: 9.72/10.0

Aug. 2025 – Sept. 2028

Online

Projects

Handy – Gesture Recognition with On-Device LLM | [\[Link\]](#) | Kotlin, MediaPipe, Gemma 3n

- Built an Android app that recognises 13 hand gestures in real time from the camera feed and translates a sequence of signs into natural English, running fully on-device with no network dependency.
- Extended MediaPipe's 7 built-in gestures with 6 custom detectors written against raw 21-point hand landmarks, adding a 4-frame stability filter to suppress false commits.
- Integrated Gemma 3n E2B (INT4) through the MediaPipe LLM Inference API in a two-pass prompt pipeline – translating gesture tokens and detecting message intent, then generating three tonally distinct reply suggestions.
- Designed the Jetpack Compose interface with a live hand-skeleton overlay, haptics and text-to-speech; resolved build-time OOM failures by moving a 2 GB model out of APK assets into runtime loading.

SafarSang – Smart Trip Planner | [\[Link\]](#) | React, Vite, Firebase, Firestore, Groq API

- Built and deployed a trip planning platform that consolidates itinerary, budgets, documents, packing and risk tracking into a single dashboard, replacing the scattered chats and spreadsheets travellers usually rely on.
- Integrated the Groq API (Llama 3.1) with a strict JSON schema and parsing layer to generate complete multi-day itineraries from a single natural-language prompt.
- Developed seven modular trip features including a Budget Splitter with expense-split calculations, a Document Vault, auto-derived Risk Flags and collaborative comments.
- Implemented Firebase authentication with email/password and Google sign-in, custom password reset and protected routing; applied Context API, custom hooks and lazy loading for state management and code-splitting.

Claribb.AI – Multi-Agent Research Workspace | [\[Link\]](#) | Next.js, TypeScript, Tailwind CSS

- Built the frontend and dashboard layer of a two-person AI research workspace that maintains persistent semantic memory across sessions and runs four specialised agents in parallel per query.
- Developed the landing and pricing pages with Framer Motion transitions and responsive layouts, plus the dashboard shell – project cards, create-project modal, research digest cards and sidebar navigation – wired to Supabase-backed project and session APIs.
- Implemented workspace panels rendering live streamed agent output over Server-Sent Events, and resolved cross-cutting lint, dependency and environment-variable issues to make the app deployment-ready on Vercel.

Skills

Programming Languages: Python, Java, JavaScript, TypeScript, SQL, HTML, CSS

Frameworks & Libraries: React, Tailwind CSS, NumPy, Pandas, Matplotlib, Seaborn

Developer Tools: Git, Postman, Firebase, Firestore, Supabase, MediaPipe, Vite, Vercel, Claude

Software: VS Code, Visual Studio, MySQL WorkBench

Achievements

Top 6 teams – Speed Run Hackathon, Scaler School of Technology.

Selected as one of 80 builders for the iQOO Hackathon 2026, a 30-hour AI build sprint held in Bengaluru.

Additional Information

Spoken Languages: English (Fluent), Hindi (Native)

Soft Skills: Team Collaboration, Communication, Analytical Thinking, Time Management, Networking